

JEPA-Based World Model for Secrecy-Aware Power Control

Concise System and Architecture Specification

Research Design Draft

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Abstract

A simplified source–HAP–UAV environment is controlled through source transmit power P_t and HAP jamming power P_j . A four-antenna HAP observes raw pilot I/Q samples, a WirelessJEPA-style encoder maps them to a compact RF latent, an MLP embeds low-dimensional network side information, an action-conditioned MDN-LSTM models latent dynamics, and a compact controller selects (P_t, P_j) . Performance is measured through AoI, AoLI, secrecy rate, secrecy outage, and power consumption. A dimension-matched VAE provides the primary representation baseline.

1 System Model

The network contains a ground source S , a legitimate HAP receiver/jammer H , and a UAV eavesdropper U . The HAP has $M_H = 4$ receive antennas. The RIS of [3] is removed.

1.1 Slot structure

Time is slotted, $t = 0, \dots, T - 1$, with

$$T_s = T_{\text{pilot}} + T_{\text{data}}. \quad (1)$$

The sequence within slot t is

pilot observation $\rightarrow$ latent encoding $\rightarrow$ power decision $\rightarrow$ data/jamming $\rightarrow$ state update.

(2)

The pilot uses fixed power P_p and the HAP jammer is silent during pilot reception.

1.2 Geometry and mobility

$$\mathbf{p}_S = [0, 0, 0]^\top, \quad (3)$$

$$\mathbf{p}_H = [0, 0, H_H]^\top, \quad (4)$$

$$\mathbf{p}_U(t) = [x_U(t), y_U(t), H_U]^\top, \quad (5)$$

$$d_{ij}(t) = \|\mathbf{p}_i(t) - \mathbf{p}_j(t)\|_2. \quad (6)$$

For a moving UAV,

$$\mathbf{p}_U(t+1) = \mathbf{p}_U(t) + T_s \mathbf{v}_U(t), \quad \mathbf{v}_U(t) = [v_x(t), v_y(t), 0]^\top. \quad (7)$$

Static, linear, random-walk, and circular trajectories may be used.

2 Wireless Channel and Noise Models

For link $i \rightarrow j$, with f_c in GHz and d_{ij} in km, use the path-loss model of [3]:

$$L_{ij}(t) [\text{dB}] = 92.45 + 20 \log_{10}(f_c) + 20 \log_{10}(d_{ij}(t)) + \alpha_{\text{atm}} d_{ij}(t) + \zeta_{ij}(t), \quad (8)$$

$$\beta_{ij}(t) = 10^{-L_{ij}(t)/10}, \quad (9)$$

$$\bar{\beta}_{ij}(t) = \beta_{ij}(t) G_i G_j \ell_i \ell_j. \quad (10)$$

The scalar fading coefficient is

$$h_{ij}(t) = \sqrt{\bar{\beta}_{ij}(t)} r_{ij}(t) e^{j\varphi_{ij}(t)}, \quad (11)$$

where r_{ij} is Nakagami- m_{ij} with $\mathbb{E}[r_{ij}^2] = 1$ and $\varphi_{ij} \sim \mathcal{U}[0, 2\pi)$.

For the $M_H = 4$ HAP ULA with spacing $d_a = \lambda/2$,

$$\mathbf{a}_H(\vartheta) = \frac{1}{\sqrt{M_H}} \begin{bmatrix} 1 & e^{-j2\pi \frac{d_a}{\lambda} \sin \vartheta} & \dots & e^{-j2\pi (M_H-1) \frac{d_a}{\lambda} \sin \vartheta} \end{bmatrix}^T, \quad (12)$$

$$\mathbf{h}_{SH}(t) = \sqrt{\bar{\beta}_{SH}(t)} \tilde{\mathbf{h}}_{SH}(t) \mathbf{a}_H(\vartheta_{SH}(t)) \in \mathbb{C}^4. \quad (13)$$

Receiver noise power is

$$\sigma_{\text{dBm}}^2 = -174 + 10 \log_{10}(B) + NF, \quad (14)$$

$$\sigma^2 = 10^{(\sigma_{\text{dBm}}^2 - 30)/10} [\text{W}]. \quad (15)$$

3 Pilot I/Q Observation

Let the normalized pilot be

$$\mathbf{s}_p = [s_p[0], \dots, s_p[W-1]], \quad \frac{1}{W} \sum_{n=0}^{W-1} |s_p[n]|^2 = 1. \quad (16)$$

For antenna $m \in \{0, \dots, 3\}$,

$$r_{m,t}[n] = \sqrt{P_p} h_{SH,m}(t) s_p[n - \tau_t] e^{j(2\pi \Delta f_t n T_{\text{samp}} + \phi_t)} + w_{m,t}[n], \quad (17)$$

where $w_{m,t}[n] \sim \mathcal{CN}(0, \sigma_p^2)$. The baseline uses $\tau_t = 0$ and $\Delta f_t = 0$.

Stacking the four complex waveforms gives

$$\mathbf{R}_t \in \mathbb{C}^{4 \times W}. \quad (18)$$

The JEPA raw input is

$$\mathbf{x}_t^{\text{IQ}}[0, m, n] = \Re\{r_{m,t}[n]\}, \quad (19)$$

$$\mathbf{x}_t^{\text{IQ}}[1, m, n] = \Im\{r_{m,t}[n]\}, \quad (20)$$

$$\boxed{\mathbf{x}_t^{\text{IQ}}} \in \mathbb{R}^{2 \times 4 \times 256} \quad (W = 256). \quad (21)$$

Per-window normalization follows WirelessJEPA [2]:

$$\mathbf{x}_t^{\text{IQ}} \leftarrow \frac{\mathbf{x}_t^{\text{IQ}}}{\max |\mathbf{x}_t^{\text{IQ}}| + \epsilon}. \quad (22)$$

4 Power-Control Environment

The controller action is

$$\mathbf{a}_t = \begin{bmatrix} P_t(t) \\ P_j(t) \end{bmatrix} \in \mathbb{R}^2, \quad 0 \leq P_t \leq P_t^{\max}, \quad 0 \leq P_j \leq P_j^{\max}. \quad (23)$$

All SINR calculations use linear power in watts.

With HAP MRC,

$$\mathbf{w}_H(t) = \frac{\mathbf{h}_{SH}(t)}{\|\mathbf{h}_{SH}(t)\|_2}, \quad (24)$$

$$\gamma_H(t) = \frac{P_t(t)\|\mathbf{h}_{SH}(t)\|_2^2}{\sigma_H^2 + \kappa P_j(t)G_{HH}(t)}, \quad (25)$$

$$\gamma_U(t) = \frac{P_t(t)|h_{SU}(t)|^2}{\sigma_U^2 + P_j(t)|h_{HU}(t)|^2}. \quad (26)$$

The rates are

$$R_H(t) = B \log_2(1 + \gamma_H(t)), \quad (27)$$

$$R_U(t) = B \log_2(1 + \gamma_U(t)), \quad (28)$$

$$\boxed{R_s(t)} = [R_H(t) - R_U(t)]^+. \quad (29)$$

For packet size L_{pkt} ,

$$R_{\text{req}} = \frac{L_{\text{pkt}}}{T_{\text{data}}}. \quad (30)$$

Define

$$g_t = \mathbb{I}\{R_s(t) \geq R_s^{\min}\}, \quad (31)$$

$$\mu_H(t) = g_t \mathbb{I}\{R_H(t) \geq R_{\text{req}}\}, \quad (32)$$

$$\mu_U(t) = g_t \mathbb{I}\{R_U(t) \geq R_U^{\text{dec}}\}. \quad (33)$$

If $g_t = 0$, confidential data are withheld/dummy-transmitted.

5 AoI, AoLI, Objective, and Reward

$$\Delta_H(t+1) = \begin{cases} 1, & \mu_H(t) = 1, \\ \min\{\Delta_H(t) + 1, \Delta_{\max}\}, & \mu_H(t) = 0, \end{cases} \quad (34)$$

$$\Delta_U(t+1) = \begin{cases} 1, & \mu_U(t) = 1, \\ \min\{\Delta_U(t) + 1, \Delta_{\max}\}, & \mu_U(t) = 0. \end{cases} \quad (35)$$

Lower Δ_H and higher Δ_U are desired.

Long-term metrics are

$$\bar{\Delta}_H = \frac{1}{T} \sum_t \Delta_H(t), \quad \bar{\Delta}_U = \frac{1}{T} \sum_t \Delta_U(t), \quad (36)$$

$$\bar{R}_s = \frac{1}{T} \sum_t R_s(t), \quad \bar{P}_t = \frac{1}{T} \sum_t P_t(t), \quad \bar{P}_j = \frac{1}{T} \sum_t P_j(t). \quad (37)$$

The control objective is

$$\boxed{\min_{\pi} \bar{\Delta}_H - \omega \bar{\Delta}_U + \lambda_t \bar{P}_t + \lambda_j \bar{P}_j} \quad (38)$$

subject to the power limits and the secrecy-gated transmission rule.
For exhaustive search and optional RL, define

$$\tilde{R}_s(t) = \min \left\{ \frac{R_s(t)}{R_s^{\text{ref}}}, 1 \right\}, \quad (39)$$

$$\tilde{P}_t(t) = P_t(t)/P_t^{\max}, \quad \tilde{P}_j(t) = P_j(t)/P_j^{\max}, \quad (40)$$

$$\begin{aligned} r_t = & -w_H \frac{\Delta_H(t+1)}{\Delta_{\max}} + w_U \frac{\Delta_U(t+1)}{\Delta_{\max}} + w_s \tilde{R}_s(t) \\ & - w_P (\tilde{P}_t(t) + \tilde{P}_j(t)) - w_v \mathbb{I}\{R_s(t) < R_s^{\min}\}. \end{aligned} \quad (41)$$

Initial reward weights:

$$(w_H, w_U, w_s, w_P, w_v) = (1, 0.2, 0.5, 0.05, 1). \quad (42)$$

6 State and Observation Model

The simulator retains privileged physical state

$$\mathbf{s}_t^{\text{env}} = \{\mathbf{p}_U, \mathbf{v}_U, \mathbf{h}_{SH}, h_{SU}, h_{HU}, h_{HH}, \Delta_H, \Delta_U\}. \quad (43)$$

The agent does not receive this state directly.

The high-dimensional observation is $\mathbf{x}_t^{\text{IQ}}$. The compact side-information vector is

$$\mathbf{q}_t = \begin{bmatrix} \tilde{\Delta}_H(t) \\ \tilde{\Delta}_U(t) \\ \tilde{\beta}_{SU}(t) \\ \tilde{\beta}_{HU}(t) \\ \tilde{v}_x(t) \\ \tilde{v}_y(t) \\ \tilde{P}_t(t-1) \\ \tilde{P}_j(t-1) \end{bmatrix} \in \mathbb{R}^8. \quad (44)$$

The large-scale eavesdropper-link gains are assumed available from tracking/sensing; exact instantaneous h_{SU} and h_{HU} remain privileged.

7 WirelessJEPA Representation Block

7.1 Input preprocessing

To match WirelessJEPA [2], repeat each antenna row 64 times:

$$\mathbf{x}_{\text{raw}} \in \mathbb{R}^{2 \times 4 \times 256} \longrightarrow \mathbf{x} \in \mathbb{R}^{2 \times 256 \times 256}, \quad (45)$$

$$x_{c,i,n} = (x_{\text{raw}})_{c, \lfloor i/64 \rfloor, n}. \quad (46)$$

Native $2 \times 4 \times 256$ processing is retained as an ablation.

7.2 JEPA pretraining

The pretraining networks are context encoder E_θ , predictor P_ϕ , and EMA target encoder $E_{\bar{\theta}}$:

$$\mathbf{H}_t = E_\theta(\mathbf{x}_t, \mathbf{M}_{\text{lat}}), \quad (47)$$

$$\hat{\mathbf{Y}}_t = P_\phi(\hat{\mathbf{H}}_t), \quad (48)$$

$$\mathbf{Y}_t = E_{\bar{\theta}}(\mathbf{x}_t). \quad (49)$$

For masked latent positions $\mathcal{M}$,

$$\mathcal{L}_{\text{JEPA}} = \frac{1}{|\mathcal{M}|} \sum_{(i,j) \in \mathcal{M}} \|\hat{\mathbf{Y}}_{t,i,j} - \mathbf{Y}_{t,i,j}\|_2^2. \quad (50)$$

The target encoder is updated by

$$\bar{\theta} \leftarrow \tau \bar{\theta} + (1 - \tau) \theta, \quad \tau : 0.996 \rightarrow 1. \quad (51)$$

Use antenna, time, and multi-block masks; multi-block is the default.

7.3 Deployment RF latent

After pretraining, discard P_ϕ and $E_{\bar{\theta}}$. For the full input,

$$\mathbf{H}_t^{\text{full}} = E_\theta(\mathbf{x}_t), \quad (52)$$

$$\boxed{\mathbf{z}_t^{\text{RF}} = W_{\text{RF}} \text{GAP}(\mathbf{H}_t^{\text{full}}) + \mathbf{b}_{\text{RF}} \in \mathbb{R}^{128}.} \quad (53)$$

8 Side-Information Encoder and Joint Latent

Use a two-layer MLP

$$\mathbf{e}_t = \text{ReLU}(W_1 \mathbf{q}_t + \mathbf{b}_1) \in \mathbb{R}^{64}, \quad (54)$$

$$\mathbf{z}_t^{\text{net}} = W_2 \mathbf{e}_t + \mathbf{b}_2 \in \mathbb{R}^{32}. \quad (55)$$

The complete latent state is

$$\boxed{\mathbf{z}_t = \begin{bmatrix} \mathbf{z}_t^{\text{RF}} \\ \mathbf{z}_t^{\text{net}} \end{bmatrix} \in \mathbb{R}^{160}.} \quad (56)$$

Thus

$$(\mathbf{x}_t^{\text{IQ}}, \mathbf{q}_t) \longrightarrow (128, 32) \longrightarrow \mathbf{z}_t \in \mathbb{R}^{160}. \quad (57)$$

9 MDN-LSTM World Model

Normalize the current action as

$$\tilde{\mathbf{a}}_t = \begin{bmatrix} P_t(t)/P_t^{\text{max}} \\ P_j(t)/P_j^{\text{max}} \end{bmatrix} \in [0, 1]^2. \quad (58)$$

The recurrent input is

$$\boxed{\mathbf{u}_t = [\mathbf{z}_t; \tilde{\mathbf{a}}_t] \in \mathbb{R}^{162}.} \quad (59)$$

The memory update is

$$\boxed{(\mathbf{h}_{t+1}, \mathbf{c}_{t+1}) = \text{LSTM}_\psi(\mathbf{u}_t, \mathbf{h}_t, \mathbf{c}_t), \quad \mathbf{h}_t, \mathbf{c}_t \in \mathbb{R}^{256}.} \quad (60)$$

Use $K = 5$ Gaussian mixtures. For latent dimension $d = 1, \dots, 160$,

$$\pi_{d,k,t} = \text{softmax}_k(\ell_{d,k,t}), \quad (61)$$

$$\sigma_{d,k,t} = \text{softplus}(s_{d,k,t}) + \sigma_{\min}, \quad (62)$$

$$p(\mathbf{z}_{t+1} | \mathbf{z}_t, \mathbf{a}_t, \mathbf{h}_t) = \prod_{d=1}^{160} \sum_{k=1}^5 \pi_{d,k,t} \mathcal{N}(\mathbf{z}_{t+1,d}; \mu_{d,k,t}, \sigma_{d,k,t}^2). \quad (63)$$

The MDN head outputs

$$160 \times 5 \times 3 = \boxed{2400} \quad (64)$$

raw mixture parameters per step. Train with

$$\mathcal{L}_{\text{MDN}} = -\frac{1}{N_{\text{seq}}} \sum_{\text{sequences}} \sum_t \log p(\mathbf{z}_{t+1} | \mathbf{z}_t, \mathbf{a}_t, \mathbf{h}_t). \quad (65)$$

Optional imagined-rollout heads are

$$\hat{r}_t = f_r(\mathbf{h}_{t+1}), \quad (66)$$

$$\hat{g}_t = \sigma(f_g(\mathbf{h}_{t+1})). \quad (67)$$

10 Controller

At decision time t , $\mathbf{h}_t$ contains information from previous transitions. The controller uses

$$\mathbf{f}_t = [\mathbf{z}_t; \mathbf{h}_t] \in \mathbb{R}^{160+256} = \mathbb{R}^{416}. \quad (68)$$

A World-Models-style linear controller is

$$\mathbf{u}_t^C = W_C \mathbf{f}_t + \mathbf{b}_C \in \mathbb{R}^2, \quad (69)$$

$$P_t(t) = P_t^{\max} \sigma(u_{t,1}^C), \quad (70)$$

$$P_j(t) = P_j^{\max} \sigma(u_{t,2}^C). \quad (71)$$

Hence

$$416 \rightarrow 2. \quad (72)$$

The chronological recursion is

$$(\mathbf{x}_t^{\text{IQ}}, \mathbf{q}_t) \rightarrow \mathbf{z}_t, \quad (73)$$

$$(\mathbf{z}_t, \mathbf{h}_t) \rightarrow \mathbf{a}_t, \quad (74)$$

$$(\mathbf{z}_t, \mathbf{a}_t, \mathbf{h}_t, \mathbf{c}_t) \rightarrow (\mathbf{h}_{t+1}, \mathbf{c}_{t+1}, p(\mathbf{z}_{t+1})), \quad (75)$$

$$(\mathbf{s}_t^{\text{env}}, \mathbf{a}_t) \rightarrow (r_t, \mathbf{x}_{t+1}^{\text{IQ}}, \mathbf{q}_{t+1}, \mathbf{s}_{t+1}^{\text{env}}). \quad (76)$$

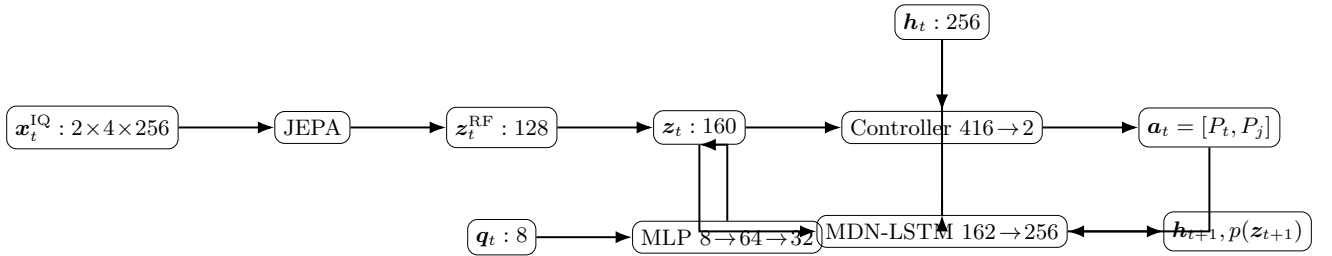


Figure 1: Deployment-time architecture and dimensions.

11 Datasets and Training

Three datasets are maintained.

11.1 JEPA pretraining dataset

$$\mathcal{D}_{\text{JEPA}} = \{\mathbf{x}_n^{\text{IQ}}\}_{n=1}^{N_{\text{JEPA}}}. \quad (77)$$

Randomize channel/AoA/SNR/shadowing/fading and optional ϕ , Δf , and τ . No action labels are required.

11.2 World-model trajectory dataset

$$\mathcal{D}_{\text{WM}} = \{(\mathbf{x}_t^{\text{IQ}}, \mathbf{q}_t, \mathbf{a}_t, r_t, \mathbf{x}_{t+1}^{\text{IQ}}, \mathbf{q}_{t+1})\}. \quad (78)$$

Use random, grid, heuristic, and later expert actions so that the transition model covers the action space.

11.3 Exhaustive-search expert dataset

Define dBm grids

$$\mathcal{P}_t^{\text{dBm}} = \{0, 1, \dots, 30\}, \quad (79)$$

$$\mathcal{P}_j^{\text{dBm}} = \{0, 1, \dots, 30\}, \quad (80)$$

$$\mathcal{A}_{\text{grid}} = \mathcal{P}_t^{\text{dBm}} \times \mathcal{P}_j^{\text{dBm}}, \quad |\mathcal{A}_{\text{grid}}| = 961. \quad (81)$$

Convert every candidate to watts before evaluating the physical equations. Clone the same simulator state for each candidate and compute

$$\mathbf{a}_t^* = \arg \max_{\mathbf{a} \in \mathcal{A}_{\text{grid}}} r_t(\mathbf{a}). \quad (82)$$

The expert trajectory dataset is

$$\mathcal{D}_{\text{ES}} = \{(\mathbf{x}_t^{\text{IQ}}, \mathbf{q}_t, \mathbf{a}_t^*, r_t^*, \mathbf{x}_{t+1}^{\text{IQ}}, \mathbf{q}_{t+1})\}. \quad (83)$$

This oracle is one-step/myopic; long-horizon optimization is handled later through CMA-ES/PPO or a short-horizon oracle.

11.4 Controller behavior cloning

With frozen JEPA and MDN-LSTM,

$$\mathcal{L}_{\text{BC}} = \frac{1}{N} \sum_t \|\tilde{\mathbf{a}}_t^C - \tilde{\mathbf{a}}_t^*\|_2^2. \quad (84)$$

The hidden state $\mathbf{h}_t$ is generated sequentially along each expert trajectory.

11.5 Training order

1. Validate the physical simulator and exhaustive-search objective.
2. Collect broad I/Q and transition trajectories.
3. Pretrain JEPA; retain E_θ and freeze it.
4. Encode trajectories and train the MDN-LSTM with teacher forcing.
5. Generate exhaustive-search expert trajectories.
6. Freeze JEPA/MDN-LSTM and behavior-clone the controller.
7. Optionally fine-tune the controller in the real simulator using CMA-ES or PPO.
8. Attempt imagined-rollout training only after validating multi-step world-model error.

12 Primary VAE Baseline

The VAE receives the same normalized I/Q input and uses the same RF latent dimension:

$$q_\varphi(\mathbf{z}^{\text{RF}}|\mathbf{x}) = \mathcal{N}(\boldsymbol{\mu}_\varphi(\mathbf{x}), \text{diag}(\boldsymbol{\sigma}_\varphi^2(\mathbf{x}))), \quad (85)$$

$$\mathcal{L}_{\text{VAE}} = \mathbb{E}_{q_\varphi} [\|\mathbf{x} - \hat{\mathbf{x}}\|_2^2] + \beta_{\text{KL}} D_{\text{KL}}(q_\varphi(\mathbf{z}|\mathbf{x})\|\mathcal{N}(0, I)), \quad (86)$$

$$d_{\text{RF,VAE}} = d_{\text{RF,JEPA}} = 128. \quad (87)$$

The side encoder, MDN-LSTM, controller, training data, exhaustive-search labels, and evaluation seeds must remain identical.

13 Reference Dimensions and Parameters

Table 1: Architecture dimensions.

Block	Input	Output
Pilot I/Q	4×256 complex	$2 \times 4 \times 256$ real
JEPA preprocessing	$2 \times 4 \times 256$	$2 \times 256 \times 256$
JEPA + projection	$2 \times 256 \times 256$	128
Side MLP	8	32 ($8 \rightarrow 64 \rightarrow 32$)
Latent fusion	$128 + 32$	160
Controller	$160 + 256 = 416$	2
MDN-LSTM input	$160 + 2$	162
LSTM state	$162 + \text{states}$	$h, c \in \mathbb{R}^{256}$
MDN head	256	2400 parameters

Table 2: Initial simulation/training values. Values not fixed by the source papers are starting points to sweep.

Parameter	Value	Note
f_c	28 GHz	source-aligned
B	20 MHz	source-aligned
T_s	1 s	source-aligned
H_H	20 km	source-aligned
Source-UAV horizontal range	5 km	initial scale
α_{atm}	0.06 dB/km	source-aligned
NF	5 dB	source-aligned
κ	0.35	source-aligned
m_{SH}, m_{SU}, m_{HU}	3, 2, 2	m_{HU} proposed
P_t, P_j	0–30 dBm	search/control range
P_p	10 dBm	proposed fixed pilot power
W	256	WirelessJEPA-aligned
L_{pkt}	2 Mbit	source-aligned frame size
Δ_{max}	200	proposed
R_s^{min}	0.5 Mbit/s	proposed; sweep
JEPA epochs	100	WirelessJEPA-aligned
JEPA optimizer	AdamW + cosine decay	WirelessJEPA-aligned
EMA τ	$0.996 \rightarrow 1$	WirelessJEPA-aligned
$d_{\text{RF}}, d_{\text{net}}, d_z$	128, 32, 160	proposed
LSTM hidden, mixtures	256, 5	World Models-aligned

14 Evaluation

Primary control metrics:

$$\bar{\Delta}_H, \quad \bar{\Delta}_U, \quad \bar{R}_s, \quad (88)$$

$$P_{\text{sec, out}} = \frac{1}{T} \sum_t \mathbb{I}\{R_s(t) < R_s^{\text{min}}\}, \quad (89)$$

$$\bar{P}_t, \quad \bar{P}_j, \quad \eta_{\text{succ}} = \frac{1}{T} \sum_t \mu_H(t), \quad \eta_{\text{leak}} = \frac{1}{T} \sum_t \mu_U(t). \quad (90)$$

World-model metrics:

$$\mathcal{L}_{\text{MDN}}^{\text{test}}, \quad \text{MSE}(\hat{z}_{t+1}, z_{t+1}), \quad H\text{-step latent rollout error}. \quad (91)$$

Controller metrics:

$$\text{MSE}(\mathbf{a}_t^C, \mathbf{a}_t^*), \quad \Pr(|P_t - P_t^*| \leq 2 \text{ dB}), \quad \Pr(|P_j - P_j^*| \leq 2 \text{ dB}). \quad (92)$$

Minimum baselines:

1. random and fixed power;
2. one-step exhaustive-search oracle;
3. engineered-state MLP;
4. VAE $\rightarrow$ MDN-LSTM $\rightarrow$ controller;
5. JEPA $\rightarrow$ controller without recurrent memory;
6. JEPA $\rightarrow$ MDN-LSTM $\rightarrow$ controller.

Required JEPA ablations: antenna/time/multi-block masking, native vs. repeated antenna dimension, $d_{\text{RF}} \in \{64, 128, 256\}$, and frozen vs. partial fine-tuning.

15 Implementation Constraints

- Convert dBm to watts before all SINR calculations.
- Use km in the path-loss equation and Hz in the noise-power equation.
- At action time, the controller may use only $(\mathbf{x}_t^{\text{IQ}}, \mathbf{q}_t, \mathbf{h}_t)$.
- Do not expose post-action $R_H(t)$, $R_U(t)$, $R_s(t)$, or exact instantaneous $h_{SU}(t)$, $h_{HU}(t)$ to the controller.
- Split train/validation/test data by trajectory/channel seed, not adjacent I/Q windows.
- Validate that increasing P_t generally raises both legitimate and eavesdropper rates, while increasing P_j suppresses the UAV but can hurt the HAP through residual self-interference.
- AoI resets only after legitimate success; AoLI resets only after UAV decoding.

16 Controlled Research Comparison

The primary experiment is

VAE $\rightarrow$ same MDN-LSTM $\rightarrow$ same controller JEPA $\rightarrow$ same MDN-LSTM $\rightarrow$ same controller (93)
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with identical environment trajectories, action space, expert labels, downstream dimensions, controller training, and test seeds.

References

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